Six Explicit Verified-Base Cone Cases

Abstract

We prove a conditional cone lemma from first principles and apply it to six explicitly verified five-vertex bases. The bases are Atlas 33, 34, 36, 40, 45, and 49. Each already has an accepted graphon lower bound, and its exact chromatic target is explicitly factorized and checked to be nonnegative, increasing, and convex from its chromatic threshold. The result proves the previously open cone graphs Atlas 177, 179, 183, 192, 200, and 205. No general universal-vertex closure is asserted.

1 A conditional cone lemma

Let $W\colon \Omega^2 \to [0,1]$ be a symmetric measurable graphon on a probability space (Ω, μ). For a finite graph G, write

$$t(G,W) = \int_{\Omega^{V(G)}} \prod_{uv\in E(G)} W(x_u, x_v) \prod_{u\in V(G)} \mathrm{d}\mu(x_u).$$

Put

$$p = t(K_2, W), \qquad d(x) = \int_\Omega W(x,y)\,\mathrm{d}\mu(y).$$

Define

$$\tau(x) = \int_{\Omega^2} W(x,y)W(x,z)W(y,z)\,\mathrm{d}\mu(y)\,\mathrm{d}\mu(z).$$

Lemma 1.1 (Weighted rooted-triangle inequality)**.** *If $p > 0$, then for every integer $h \geq 0$,*

$$\int_\Omega d(x)^h \tau(x)\,\mathrm{d}\mu(x) \geq \frac{2p-1}{p} \int_\Omega d(x)^{h+2}\,\mathrm{d}\mu(x).$$

Proof. Put $M_j = \int d^j$, let T_W be the graphon operator, and write

$$I_h = \int d^h \tau, \qquad J_h = \left\langle d^h, T_W d \right\rangle.$$

Applying $ab \geq a + b - 1$ to $W(x,y), W(x,z)$, multiplying by $d(x)^h W(y,z)$, and integrating gives

$$I_h \geq 2J_h - pM_h. \tag{1}$$

Set $U = 1 - W$, $u = T_U \mathbf{1} = 1 - d$, and $q = 1 - p$. Then

$$T_W d = d - q\mathbf{1} + T_U u,$$

so

$$J_h = M_{h+1} - qM_h + \left\langle d^h, T_U u \right\rangle. \tag{2}$$

By symmetry of U, twice

$$\left\langle d^h, T_U u \right\rangle - \int d^h u^2$$

equals

$$\int_{\Omega^2} U(x,y)\big(u(y)-u(x)\big)\big(d(x)^h - d(y)^h\big)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y) \geq 0,$$

because $d = 1 - u$. Therefore (2) gives

$$J_h \geq M_{h+2} - M_{h+1} + pM_h.$$

Using this in (1), and then subtracting $(2p-1)M_{h+2}/p$, leaves exactly

$$\frac{1}{p}\int_\Omega d(x)^h (d(x)-p)^2\,\mathrm{d}\mu(x) \geq 0.$$

□

For $d(x) > 0$, define

$$\mathrm{d}\mu_x(y) = \frac{W(x,y)}{d(x)}\,\mathrm{d}\mu(y),$$

and let W_x denote W on (Ω, μ_x). Its edge density is $z_x = \tau(x)/d(x)^2$. Put $z_x = 0$ when $d(x) = 0$.

Corollary 1.2 (Weighted link average)**.** *For every integer $N \geq 2$,*

$$\int_\Omega d(x)^N z_x\,\mathrm{d}\mu(x) \geq \left(2 - \frac{1}{p}\right)\int_\Omega d(x)^N\,\mathrm{d}\mu(x).$$

Proof. Apply Lemma 1.1 with $h = N - 2$. □

Lemma 1.3 (Verified-base cone lemma)**.** *Let F be a graph on $N \geq 2$ vertices and let*

$$\phi(z) = \Phi_F(z).$$

Assume that for some $a \in [0,1)$:

1. *every graphon V of edge density $z \geq a$ satisfies $t(F,V) \geq \phi(z)$;*
2. *on $[a,1]$, one has $\phi \geq 0$, $\phi' \geq 0$, and $\phi'' \geq 0$;*
3. *$\phi(a) = 0$.*

Then every graphon W of edge density

$$p \geq b := \frac{1}{2-a}$$

satisfies

$$t(K_1 \vee F, W) \geq \Phi_{K_1 \vee F}(p).$$

Proof. Put $c = 2 - 1/p$. The assumption $p \geq 1/(2-a)$ is exactly $c \geq a$. Convexity shows that every graphon V of arbitrary edge density z satisfies the tangent bound

$$t(F, V) \geq \phi(c) + \phi'(c)(z - c).$$

Indeed, for $z \geq a$ this follows from the assumed graphon bound and convexity. For $z < a$, the tangent is nonpositive because it is at most $\phi(a) = 0$ at a and has nonnegative slope, whereas $t(F, V) \geq 0$.

Conditioning on the image x of the new cone vertex gives

$$t(K_1 \vee F, W) = \int_\Omega d(x)^N t(F, W_x)\, \mathrm{d}\mu(x).$$

Let $M_N = \int d^N$. Applying the tangent bound in every link,

$$\begin{aligned} t(K_1 \vee F, W) \geq\ & M_N \phi(c) \\ & + \phi'(c) \left(\int d(x)^N z_x \, \mathrm{d}\mu(x) - c M_N \right). \end{aligned}$$

The second line is nonnegative by Corollary 1.2. Jensen then gives

$$t(K_1 \vee F, W) \geq p^N \phi(c).$$

If $q = 1 - p$, then $1 - c = q/p$. Since $\chi_{K_1 \vee F}(x) = x \chi_F(x - 1)$,

$$\begin{aligned} \Phi_{K_1 \vee F}(p) &= q^N \chi_F\left(\frac{p}{q}\right) \\ &= p^N \left(\frac{q}{p}\right)^N \chi_F\left(\frac{p}{q}\right) = p^N \phi(c). \end{aligned}$$

□

2 The six verified bases

Let F_j denote NetworkX Atlas graph j. The following table gives the complete input verification. The graphon bound column records the accepted reason already present in the catalogue before this theorem.

j	graph6	accepted bound	a	$\phi_j(z) = \Phi_{F_j}(z)$
33	`Dn?`	diamond plus isolate product	1/2	$z(2z-1)^2$
34	`D@{`	rooted common leaves	1/2	$z^3(2z-1)$
36	`DJc`	rooted two-edge tail	1/2	$z^3(2z-1)$
40	`DjW`	forest-cone theorem	1/2	$z^2(2z-1)^2$
45	`DJ{`	clique common leaf	2/3	$z^2(2z-1)(3z-2)$
49	`Df{`	special paw cone	2/3	$z(2z-1)^2(3z-2)$

The chromatic polynomials, obtained either from the displayed structures or by deletion–contraction, are

$$\begin{aligned} \chi_{F_{33}}(x) &= x^2(x-1)(x-2)^2, \\ \chi_{F_{34}}(x) = \chi_{F_{36}}(x) &= x(x-1)^3(x-2), \\ \chi_{F_{40}}(x) &= x(x-1)^2(x-2)^2, \\ \chi_{F_{45}}(x) &= x(x-1)^2(x-2)(x-3), \\ \chi_{F_{49}}(x) &= x(x-1)(x-2)^2(x-3). \end{aligned}$$

Substitution in the definition of Φ gives exactly the table.

For completeness, the required shape conditions are also checked explicitly. For the first four rows, every displayed factor is nonnegative and nondecreasing on $[1/2, 1]$; for the last two, on $[2/3, 1]$. A product of nonnegative nondecreasing affine factors is nonnegative, nondecreasing, and convex: its first and second derivatives are sums of products of nonnegative factors and positive constant slopes. Repeated factors cause no problem. Each polynomial vanishes at its displayed a. Thus every hypothesis of Lemma 1.3 is verified, not inferred from tests.

3 New positive cases

Theorem 3.1 (Six explicit cones)**.** *For*

$$j \in \{33, 34, 36, 40, 45, 49\},$$

the cone $K_1 \vee F_j$ satisfies

$$t(K_1 \vee F_j, W) \geq \Phi_{K_1 \vee F_j}(p)$$

for every graphon throughout its full required density interval.

Proof. For $j = 33, 34, 36, 40$, the verified-base threshold is $a = 1/2$, so Lemma 1.3 gives the cone threshold

$$b = \frac{1}{2 - 1/2} = \frac{2}{3}.$$

Each base is 3-chromatic, so each cone is 4-chromatic and this is the exact required threshold.

For $j = 45, 49$, one has $a = 2/3$, and the cone threshold is

$$b = \frac{1}{2 - 2/3} = \frac{3}{4}.$$

Each base is 4-chromatic, so each cone is 5-chromatic and this is again the exact required threshold. □

Corollary 3.2 (Catalogue IDs)**.** *The newly positive Atlas graphs are*

Atlas ID	*graph6*	*base Atlas*	*threshold*
177	`En{G`	33	2/3
179	`E_~w`	34	2/3
183	`EjvG`	36	2/3
192	`E~Xo`	40	2/3
200	`E~{W`	45	3/4
205	`E~~G`	49	3/4

At the threshold the target vanishes, and the balanced complete $(\chi(H) - 1)$-partite graphon attains equality. At $p = 1$, both sides equal 1. At every balanced multipartite grid point in between, equality follows from the chromatic-polynomial identity.

4 Scope and hidden assumptions

The proof applies Lemma 1.3 only to the six bases whose exact graphon inequalities and analytic target properties were verified in the table. It does not treat arbitrary positive bases and does not assert a general universal-vertex or join closure.

Individual graphon links may have density below the base threshold; the tangent proof handles those links by a nonpositive lower bound. The argument uses ordinary homomorphism densities directly and assumes no regularity, step structure, injectivity, minimum degree, or pointwise degree lower bound.

A What the Lean formalization actually proves

Theorem 3.1 is formalized for all six cones; Atlas 177, 179, 183, 192, 200 and 205 are `verified`. The development is `lean/Taeyoung/Methods/BaseCone/`, over the abstract `Methods/ConeBound.lean`; Atlas 192 is in `Methods/ForestCone/Rows.lean` instead, because its base bound is the forest-cone theorem. One difference of substance.

Hypothesis (2) of Lemma 1.3 is replaced, not verified

The convexity hypothesis $\phi \geq 0$, $\phi' \geq 0$, $\phi'' \geq 0$ on $[a, 1]$ is not checked for the six targets. It is replaced by an algebraic one: each Φ_{F_j} is a product of affine factors

$$\phi(z) = \prod_i \big(1 - k_i(1 - z)\big), \qquad k_i \geq 0,$$

and the cone lemma is stated in Lean for that class (`satisfiesLowerBound_of_baseCone`). The six applications then reduce to naming a list of slopes k_i, with no second derivative and no convexity theory anywhere in the development.

The affine form also improves the last step. Lemma 1.3 bounds $p^N \phi(2 - 1/p)$ below by $\Phi_{K_1 \vee F}(p)$; in the affine form the two are *equal*, which is `pow_mul_affineProd_shift`, so the cone's own target is reached exactly rather than through an estimate.

One base needs no separate input: Atlas 33 is the diamond together with an isolated vertex, which is itself the cone over $K_{1,2}$ with target z^2, so the same theorem supplies its bound.